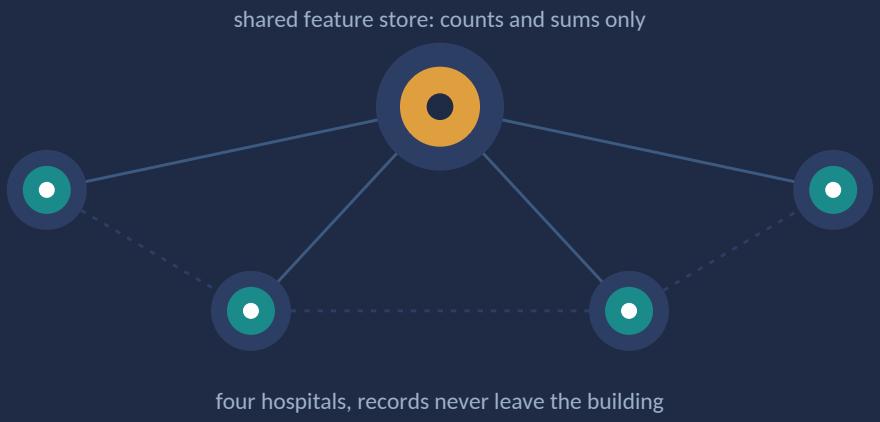
RESEARCH PROJECT | NOVEMBER 2026

Learning together, privately

Can four hospitals share one heart-disease model without sharing a single patient record, keep it private, and keep it up to date as medicine changes?



920
patients across four real hospital datasets

+8.4 pts
accuracy from one shared feature store, in 10 of 10 runs

33 to 10
points forgotten after drift: unprotected versus replay

Harshita Pandita and Pranjal Tomar

A simulation study on public data (UCI Heart Disease). Not clinically validated.
Code, results and report: github.com/hersheyharshi/federated-feature-store

AT A GLANCE

The project in six numbers

Four hospital datasets, one shared model, and a plain account of what worked and what did not. Teal numbers are good news, coral ones are the honest bad news.

81.0%
federated accuracy
average over four hospitals, with a shared feature store

+8.4
points from shared definitions
95% interval 7.4 to 9.4, better in 10 of 10 runs

0.1
points below the pooled ceiling
pooling all records is not allowed in real life

2.7x
epsilon 1 privacy limit
any one patient changes any outcome by at most this factor

33 to 10
points forgotten
unprotected versus replay, after simulated drift

3 of 10
gentle drift caught
alarm is good on strong drift (8 of 10), weak on gentle

How to read this document
The next page explains the question and the data, then the pipeline, then the results. The last page says what we can and cannot claim, and who did what.

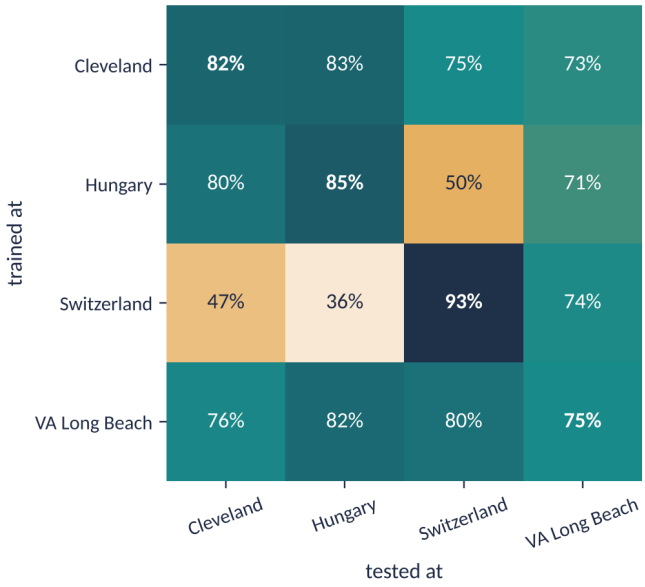
Everything was run on public data (920 patients from four sites), ten times with different random splits, and every comparison reports a 95% interval. The code, tests and result tables are in the repository, so any number here can be checked.

THE QUESTION

Four hospitals, one problem

Hospitals hold records that could teach a computer to spot heart disease earlier. They are not allowed to hand those records to each other, so each one trains on its own patients alone. That works at home and falls apart elsewhere, because a model that has only met one hospital's patients learns that hospital's habits.

Hospital	Patients	Share with heart disease	Worth knowing
Cleveland	303	46%	Almost complete records
Hungary	294	36%	Most vessel and thalassemia results missing
Switzerland	123	93%	No cholesterol values recorded at all
VA Long Beach	200	74%	About a quarter of vital signs missing



A helper trained at one hospital and tested at each of the four. Rows are where it learned, columns where it was tested. The Switzerland helper is right 47% of the time at Cleveland and 36% at Hungary, below a coin flip, because it assumes nearly everyone is sick.

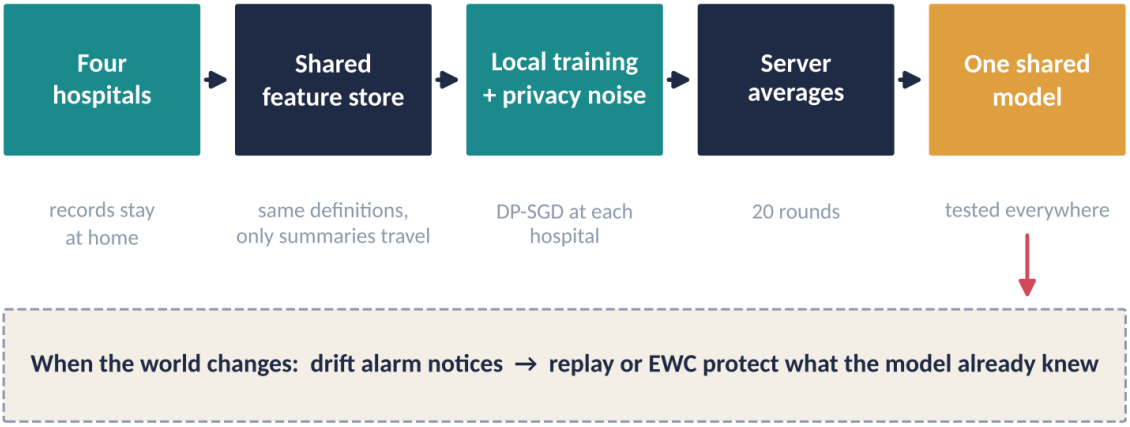
What we set out to answer

Can the four end up with one shared, sharper helper while the records never leave the building, while nobody can be picked out of the results, and while the helper keeps working as medicine changes?

WHAT WE BUILT

One pipeline, four ideas

We simulated the four hospitals from the public UCI Heart Disease data and wrote the whole pipeline ourselves in Python, so every step can be read, tested and explained.



Federated learning

Each hospital trains on its own patients and sends back only the model's weights. A server averages them, weighted by how many patients each hospital has, and sends the blend back. We repeat this 20 times.

A shared feature store

One agreed list of the 13 inputs, with meaning, unit, valid range and version. Hospitals share only counts, sums and sums of squares, from which the server works out shared averages and spreads. Records stay put.

Differential privacy

Each hospital clips how much any one patient can move the model and adds measured noise. A privacy accountant adds up the cost over all steps, and we checked ours against a published reference value. The dial is called epsilon: smaller means more privacy and more noise.

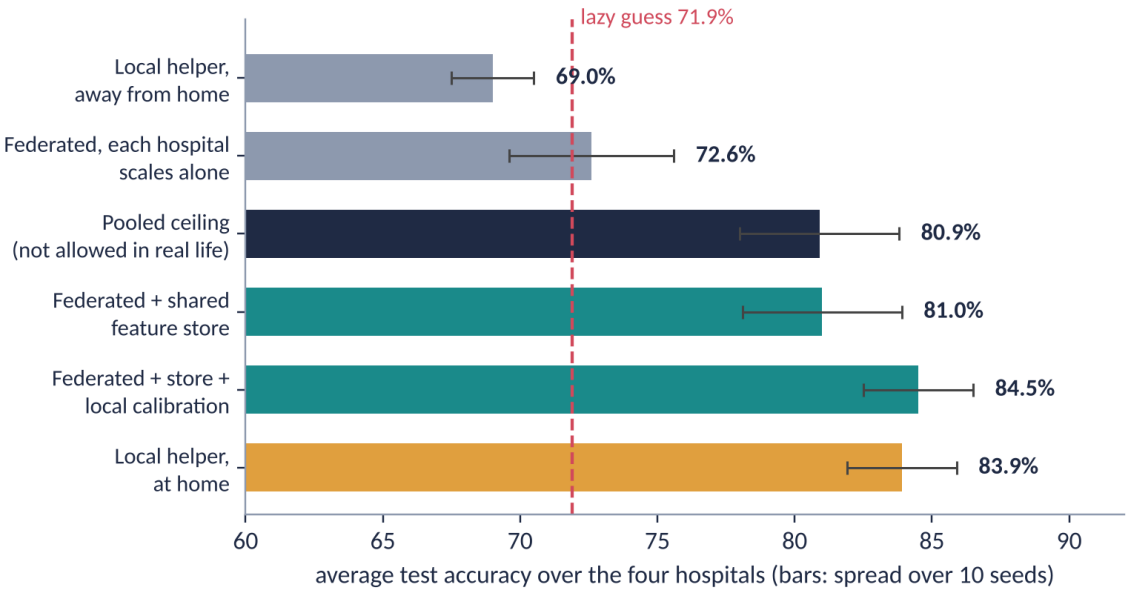
Drift and memory

We simulate a new treatment that changes who recovers, add an alarm that watches for rising mistakes, measure how much the model forgets, and compare two ways of protecting what it knew (replay and EWC).

Everything was run on two models, a logistic regression and a small neural network, over ten random seeds.

RESULTS

Learning together works, with a catch



Average test accuracy over the four hospitals, logistic regression, ten seeds. Grey bars are what happens without sharing or without agreed definitions.

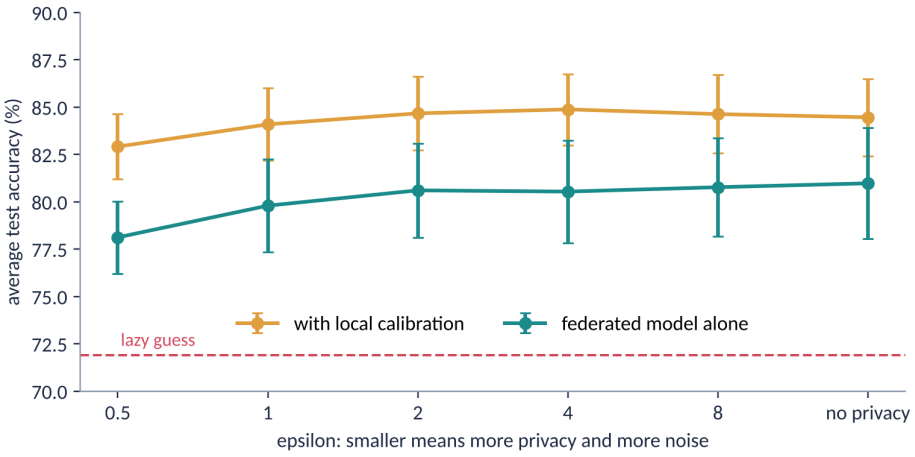
- **The shared feature store matters most.** Letting each hospital scale its own inputs leaves the federated model no better than guessing the common answer (72.6% against 71.9%). With shared definitions it reaches 81.0%, a gain of 8.4 points (95% interval 7.4 to 9.4), better in all ten runs and again with the neural network.
- **Federated learning matches the pooled ceiling.** The difference from pooling everything, which is illegal in real life, is 0.1 points (interval -0.7 to 0.9).
- **One model alone loses to a local one at Switzerland.** A single global model cannot learn that site's 93% base rate: 77.3% against 93.2%. Giving each hospital one offset, fitted on its own patients and never shared, fixes it (+15.7 points, interval 10.9 to 20.5) and brings federated learning level with local models on average.

Where we have to be careful

Even after calibration, Switzerland scores 93.0%, below the 94.6% you get by always saying "sick". The offset matches the hospital's base rate; it does not teach the model to rank those patients, and that test set is only about 37 people. We report it as a fix for the base-rate gap, not as better medicine.

RESULTS

Privacy, drift and forgetting

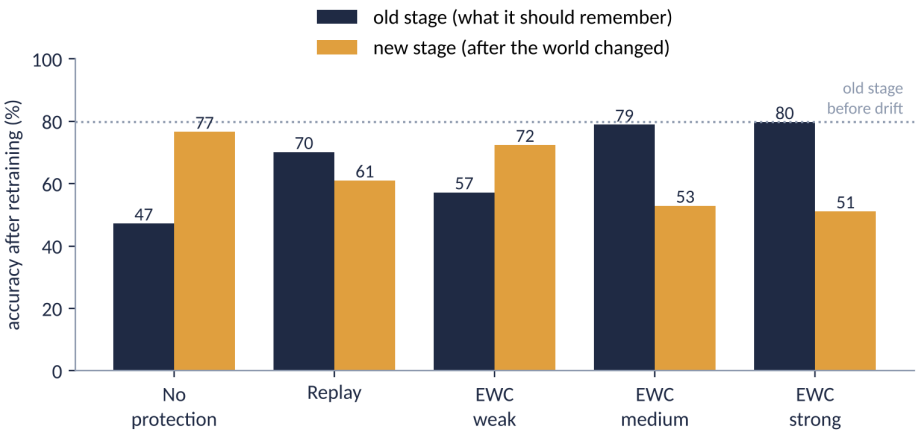


Accuracy against the privacy dial. Bars show the spread over ten seeds.

Privacy is cheap at moderate settings. With logistic regression we cannot tell epsilon 2 or more from no privacy. Epsilon 1 costs 1.2 points and epsilon 0.5 costs 2.9. The neural network pays more (3.0 and 5.5 points).

A result that did not work

We also tried making the shared summaries private with added noise. Accuracy fell to 58-62%, below the lazy guess, unless the privacy budget was so large (100) that it protected almost nothing. So we make no claim of a private feature store, and our privacy guarantee covers training only.



After the world changes, the model keeps training on new patients. Navy: how well it still does on the old patients. Amber: how well it does on the new ones.

Forgetting is real, and protection is a trade. Unprotected, the model loses 33 points on the old stage. Replay cuts that to 10 and still learns the new rule (61%). Strong EWC forgets almost nothing, but only because it barely moves: its new-stage accuracy stays near 51%. Our drift alarm caught the strong drift in 8 of 10 runs, but the gentle drift in only 3 of 10.

HONESTLY

What this is, and is not

What we can say	What we should not say
The training privacy level is a proven guarantee under stated assumptions (standard privacy accounting).	That nobody could ever be identified under any attack.
Forgetting, recovery and alarm delay are measured on drift we simulated.	That our drift bound proves the system is safe in real hospitals.
Results come from public heart data split by site to behave like separate hospitals.	That the model is ready for clinical use.

Limits worth stating

- Public data split by site, not real hospital streams; the drift and its size were chosen by us.
- Small models on one tabular dataset. Three of the four sites are missing most of two inputs.
- Ten seeds re-split the same 920 patients, so intervals reflect training and split randomness, not new patients.
- Replay and EWC were tested without privacy, so private-and-continual together is untested.

Who did what

	Responsibility
Harshita Pandita	Federation and privacy track: the federated training loop, the privacy dial and the privacy guarantee.
Pranjal Tomar	Change and memory track: simulated drift, the drift alarm, forgetting measurements and the protections.
Together	Baselines, the shared feature store, the theory notes, the report and the repository.

Reproduce it

pip install -r requirements.txt
python -m fedfs.run_all --raw data/raw --out results --download

Repository: github.com/hersheyharshi/federated-feature-store. ORCID: Harshita Pandita 0009-0002-4479-0473, Pranjal Tomar 0009-0004-6638-2424.